Nazorat testi · *Aralash test · 6-variant* — *Milliy sertifikat 2026. Kimyo*

Vaqt: *180 daqiqa*

Davriy jadvaldan foydalanish mumkin.

1. Quyidagi moddalarning qaysilarida ion bog‘ mavjud? 1) $(NH_4)_2SO_4$; 2) NH_4Cl; 3) H_2O; 4) NaCl

A) *1, 2, 3* B) *3, 4* C) *1, 2, 4* D) *1, 3*

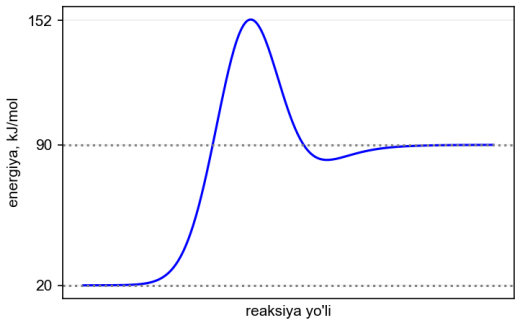
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2. 10 l N_2 va 45 l H_2 aralashmasi reaksiyaga kirishdi ($N_2 + 3H_2 \rightarrow 2NH_3$, hamma moddalar bir xil sharoitda gaz holida). Reaksiyadan keyingi gaz aralashmasining **hajmini (l) aniqlang.**

A) *55* B) *20* C) *15* D) *35*

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3. Reaksiyaning energetik diagrammasi berilgan (boshlang‘ich moddalar energiyasi 20, mahsulotlar 90 kJ/mol).



Reaksiya qaysi turga kiradi?

A) *endotermik* B) *ekzotermik*

C) *katalitik* D) *teskarilanmas*

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4. $CH_2{=}CH{-}CH{=}CH_2$ molekulasidagi σ va π bog‘lar **sonini aniqlang.**

A) *8 va 2* B) *9 va 3* C) *10 va 2* D) *9 va 2*

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5. Jadvalda Cl elementining birikmalaridagi oksidlanish darajasi berilgan:

Modda	Oksidlanish darajasi
$KClO_3$	+5
HCl	-1
NaCl	-1
HClO	+1

Ushbu moddalardan nechtasida Cl elementi ham oksidlovchi, ham qaytaruvchi xossani namoyon qiladi? (Cl uchun eng past oksidlanish darajasi -1, eng yuqorisi +7)

A) *3* B) *2* C) *4* D) *5*

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6. 40 l H_2S va 180 l O_2 aralashmasi $2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O$ tenglama bo‘yicha to‘liq reaksiyaga kirishdi (reaksiyadan so‘ng suv suyuq holatda; gazlar bir xil sharoitda o‘lchandi). Reaksiyadan so‘ng hosil bo‘lgan gazlar aralashmasining umumiy **hajmini (l) aniqlang.**

A) *220* B) *160* C) *40* D) *120*

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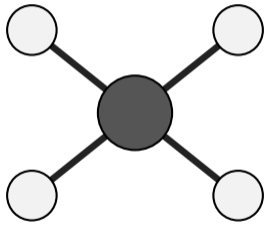
7. Na_2SO_4 ning suvli eritmasi inert elektrodlar bilan elektroliz qilinganda qaysi moddalar ajraladi?

A) *H_2 va O_2* B) *H_2, Cl_2 va NaOH*

C) *Cu va Cl_2* D) *Ag va O_2*

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8. Rasmda molekulaning sharsimon modeli tasvirlangan (qora – C, qizil – O, ko‘k – N, oq – H).



Molekuladagi π-bog‘lar **sonini aniqlang.**

A) *1* B) *0* C) *2* D) *3*

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9. Jadvalda metall plastinkalarning tuz eritmalariga tushirilishidagi kuzatuvlar berilgan (“+” – metall ajraladi, reaksiya boradi; “−” – reaksiya bormaydi):

Metall	$CuSO_4$ eritmasi	$ZnSO_4$ eritmasi	$FeSO_4$ eritmasi
Zn	+	−	+
Mg	+	+	+
Cu	−	−	−

Ushbu metallardan qaysi biri eng faol?

A) *Zn* B) *Cu* C) *Fe* D) *Mg*

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10. 4,8 g metall kislotada eritilganda 4,48 l (n.sh.) vodorod ajraldi. Metallning ekvivalent **massasini (g/mol) aniqlang** ($V_E(H_2) = 11{,}2$ l/mol).

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A) *24* B) *6* C) *53,76* D) *12*

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11. Quyidagi zarrachalardan qaysilari neon atomi bilan izoelektron? 1) Na^+; 2) Li^+; 3) Mg^{2+}; 4) F^-
A) *1, 2, 4* B) *1, 2* C) *1, 2, 3* D) *1, 3, 4*

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12. Oltingugurt ($Z = 16$) atomining tashqi energetik pog‘onasidagi elektronlar **sonini aniqlang.**
A) *7* B) *6* C) *5* D) *8*

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13. Hajmi 5 l bo‘lgan idishda A modda miqdori 100 s davomida 1,2 mol ga kamaydi. Reaksiyaning o‘rtacha tezligini (mol/(l·s)) **toping.**
A) *0,0024* B) *0,012* C) *0,24* D) *0,06*

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14. Qaysi juftliklar o‘zaro reaksiyaga kirishadi? 1) SiO_2 va H_2O; 2) Ag va H_2SO_4(suyult.); 3) $MgCl_2$ va KNO_3; 4) Fe va HCl
A) *2, 3* B) *1* C) *4* D) *3, 4*

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15. 37,8 g $Zn(NO_3)_2$ to‘liq parchalanguncha qizdirildi. Qattiq qoldiq **massasini (g) aniqlang.**
A) *37,8* B) *32,4* C) *16,2* D) *8,1*

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16. 0,2 mol NaOH tutgan eritmadan 0,15 mol CO_2 o‘tkazildi. Hosil bo‘lgan $NaHCO_3$ **massasini (g) aniqlang.**
A) *8,4* B) *2,1* C) *8* D) *4,2*

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17. Reaksiya tenglamasi: $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$. Reaksiyaga 196 g H_2SO_4 kirishdi. HCl **massasini (g) hisoblang.**
A) *146* B) *36,5* C) *292* D) *73*

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18. Li va K dan iborat 6 g aralashma suv bilan reaksiyaga kirishdi. Ajralgan vodorodning **hajmini (l, n.sh.) aniqlang.**
A) *8,96* B) *2,24* C) *4,48* D) *3*

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19. Alyuminiy (0,2 mol) va magniy (0,2 mol) aralashmasi mo‘l xlorid kislotada eritildi. Ajralgan vodorodning **hajmini (l, n.sh.) aniqlang.**
A) *11,2* B) *8,96* C) *22,4* D) *5,6*

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20. $CH_3–C(CH_3)_2–OH$ birikmaning sistematik nomi qaysi qatorda to‘g‘ri keltirilgan?
A) *butanol-2* B) *2-metilpropanol-1*
C) *2-metilpropanol-2* D) *butanol-1*

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21. Alken, alkadiyen va alkinlar haqida qaysi fikrlar to‘g‘ri? 1) Etilen bromli suvni rangsizlantiradi; 2) Kauchuk izoprenning polimeri; 3) Alkenlar $KMnO_4$ ning suvli eritmasi bilan glikol hosil qiladi; 4) Propenga HBr birikishi Markovnikov qoidasiga bo‘ysunadi
A) *1, 2, 4* B) *1, 3, 4*
C) *1, 2, 3, 4* D) *1, 2, 3*

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22. 0,6 mol asetilendan Zelinskiy usuli bilan (80% unum) benzol olindi. Benzol **massasini (g) aniqlang.**
A) *12,48* B) *37,44* C) *15,6* D) *5,2*

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23. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ jarayoni uchun 90 g $C_6H_{12}O_6$ olingan. C_2H_5OH **massasini (g) aniqlang.**
A) *46* B) *11,5* C) *92* D) *23*

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24. 11,6 g propanal Ag_2O ning ammiakli eritmasi bilan to‘liq oksidlanganda hosil bo‘lgan kumush **massasini (g) aniqlang.**
A) *21,6* B) *86,4* C) *43,2* D) *11,6*

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25. Karbon kislotalar haqida qaysi fikrlar to‘g‘ri? 1) Sirka kislota karbonatlar bilan CO_2 ajratadi; 2) Chumoli kislota "kumush ko‘zgu" reaksiyasini beradi; 3) Stearin kislota to‘yinmagan kislota; 4) Sirka angidrid ikki molekula sirka kislotadan suv chiqib hosil bo‘ladi
A) *1, 4* B) *1, 2, 3* C) *1, 3, 4* D) *1, 2, 4*

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26. 176,8 g triolein (M = 884) nikel katalizatori ishtirokida to‘liq gidrogenlandi. Sarflangan vodorod **hajmini (l, n.sh.) aniqlang.**
A) *4,48* B) *13,44* C) *26,88* D) *6,72*

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27. Uglevodlar haqida qaysi fikrlar to‘g‘ri? 1) Fruktoza ketospirt; 2) Kraxmal va selluloza bir xil M ga ega; 3) Glyukoza gidrolizlanadi; 4) Saxaroza Feling suyuqligini qaytaradi
A) *1, 3* B) *1* C) *3* D) *3, 4*

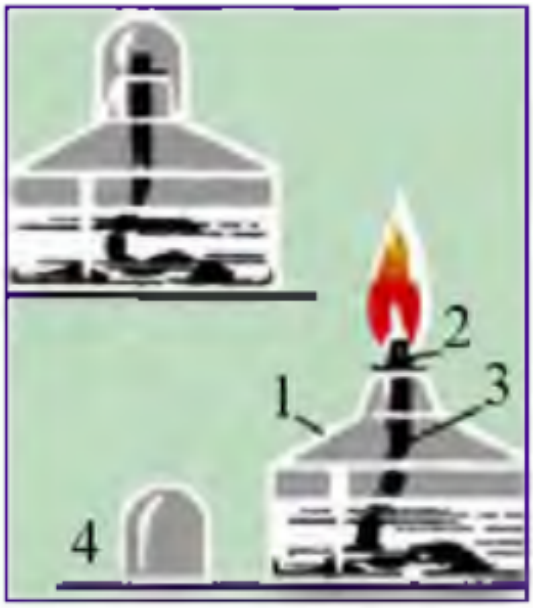
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28. 24,6 g $C_6H_5NO_2$ dan quyidagi ikki bosqichda $C_6H_5NH_3Cl$ olinadi: 1) $C_6H_5NO_2 + 3H_2 \rightarrow C_6H_5NH_2 + 2H_2O$ (unum 50%); 2) $C_6H_5NH_2 + HCl \rightarrow C_6H_5NH_3Cl$ (unum 75%). Oxirgi mahsulot $C_6H_5NH_3Cl$ ning **massasini (g) toping.**
A) *19,43* B) *9,71* C) *12,95* D) *25,9*

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29. $C_2H_2 + H_2 \rightarrow C_2H_4$ reaksiyasida 8,96 l (n.sh.) C_2H_2 ishtirok etdi. C_2H_4 **hajmini (l, n.sh.) toping.**
A) *8,96* B) *17,92* C) *4,48* D) *13,44*

30. Rasmda spirt lampasining tuzilishi ko‘rsatilgan. 3 raqami bilan **belgilang**an qism nima deb ataladi?



A) *metall disk* B) *qalpoqcha*
C) *shtativ* D) *pilik*

31. Ba^{2+} ionini aniqlash uchun qaysi reagent mos va qanday belgi kuzatiladi? A) *NaOH: qo‘ng‘ir cho‘kma ($Fe(OH)_3$)*
B) *H_2SO_4: oq cho‘kma ($BaSO_4$), HCl da erimaydi*
C) *$BaCl_2$: oq cho‘kma ($BaSO_4$), HCl da erimaydi*
D) *NaOH: oq-yashil cho‘kma ($Fe(OH)_2$)*

32. 300 g 10% li eritma tayyorlash uchun 80% li eritmadan necha gramm olish kerak (qolgani suv)?
A) *262,5* B) *30* C) *37,5* D) *75* —

Topshiriqlar (33–35) va javob variant (A–F)larini o‘zaro to‘g‘ri moslashtiring. CuO va Fe_2O_3 dan iborat 12 g aralashma qizdirilib, H_2 oqimida to‘liq qaytarildi; qoldiq (metallar) massasi 8,8 g.	A *1,8* B *3,6* C *5,4* D *6* E *5,38* F *4*
33. Aralashmadagi CuO massasini (g) aniqlang.	
34. Qaytarish uchun (20% ortiqcha olingan) sarflangan H_2 hajmini (l, n.sh.) aniqlang.	
35. Qaytarish natijasida hosil bo‘lgan suv massasini (g) aniqlang.	

36. $FeSO_4 + KMnO_4 + H_2SO_4 \rightarrow Fe_2(SO_4)_3 + MnSO_4 + K_2SO_4 + H_2O$ reaksiya tenglamasini tenglang. Reagentlar (chap qism) koeffitsiyentlari **yig‘indisini toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

37. 6,5 g Zn reaksiyaga kirishdi ($Zn + 2HCl \rightarrow ZnCl_2 + H_2$). Ajralgan H_2 **hajmini (l, n.sh.) hisoblang** (unum 75%).

Javob: _____

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Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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38. Alyuminiy (0,1 mol) va kalsiy (0,1 mol) aralashmasi mo‘l xlorid kislotada eritildi. Ajralgan vodorodning **hajmini (l, n.sh.) aniqlang.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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39. Etilatsetat va metilatsetat dan iborat 16,2 g aralashma mo‘l NaOH eritmasi bilan to‘liq gidrolizlandi. Hosil bo‘lgan spirtlar umumiy **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

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40. 12 g sirka kislota va 13,8 g etanol dan 75% unum bilan murakkab efir olindi. Efir **massasini (g) toping.**

Javob: _____

Diqqat! Javobingizni javoblar varaqasiga ko‘chirib yozing.

YOZMA ISH

41. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

$CuSO_4 \cdot 5H_2O$ va $Na_2SO_4 \cdot 10H_2O$ dan iborat 89,4 g aralashma 300 g suvda eritildi. Eritma inert elektrodlar yordamida 5 A tok kuchi bilan 3860 s elektroliz qilindi; shu vaqt tugaganda eritmadagi Cu^{2+} ionlari to‘liq tugagan edi.

1)Aralashmadagi kristallogidratlarning massalarini (g) aniqlang.

2)Dastlabki eritmada $CuSO_4$ va Na_2SO_4 ning massa ulushlarini (%) toping. Elektroliz davomida o‘tgan zaryadni (C) ham hisoblang.

3)Anodda ajralgan gazning hajmini (l, n.sh.) va elektroliz tugagandan keyingi eritma massasini (g) aniqlang.

4)Elektroliz tugagach eritmadagi H_2SO_4 va Na_2SO_4 ning massa ulushlarini (%) hisoblang. Dastlabki $CuSO_4$ ning massasini (g) ham hisoblang.

5)Shu eritmani to‘liq neytrallash uchun zichligi 1,22 g/ml bo‘lgan 20% li NaOH eritmasidan necha ml kerak? Agar tok kuchi 2 marta oshirilsa, xuddi shu miqdor Cu ajralishi uchun necha sekund kerak bo‘ladi?

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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42. Reaksiya tenglamalarini tugallang va tenglang. Hosil bo‘lgan organik moddalarning tuzilish formulalarini yozing va ularni nomlang.

1)$CH_3–CH(OH)–CH_3 + CuO \xrightarrow{t}$

2)$CH_3COOCH_3 + NaOH \xrightarrow{t}$

3)$C_6H_{12}O_6 \xrightarrow{\text{spirtli bijg‘ish}}$

4)$CH_3–C{\equiv}CH + H_2O \xrightarrow{Hg^{2+}}$

5)$3C_2H_2 \xrightarrow{C,\ t}$

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6)$C_6H_6 + HNO_3 \xrightarrow{H_2SO_4}$
7)n $CH_2{=}CH{-}CH{=}CH_2 \rightarrow$

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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43. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.

NaBr va NaI dan iborat 17,8 g aralashma suvda eritilib, mo‘l xlor o‘tkazildi. Eritma bug‘latilib, qoldiq quritilganda uning massasi 8,78 g bo‘ldi.
1)Reaksiya tenglamalarini yozing. Aralashmadagi NaBr va NaI massalarini (g) toping.
2)Sarflangan xlor hajmini (l, n.sh.) hisoblang. Hosil bo‘lgan NaCl miqdorini (mol) ham toping.
3)Ajralgan brom va yodning massalarini (g) toping.
4)Dastlabki aralashma eritmasiga mo‘l $AgNO_3$ qo‘shilganda hosil bo‘ladigan cho‘kmalar massasini (g) aniqlang.
5)Ajralgan brom va yod aralashmasida bromning massa ulushini (%) toping.

Diqqat! Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

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Nazorat testi · *6-variant* **Javoblar kaliti**

Faqat o‘qituvchi uchun.

1–32-savollar

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
C	*D*	*A*	*D*	*B*	*B*	*A*	*B*	*D*	*D*	*D*	*B*	*A*	*C*	*C*	*D*
17	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**
A	*C*	*A*	*C*	*C*	*A*	*A*	*C*	*D*	*B*	*B*	*B*	*A*	*D*	*B*	*C*

33–35-savollar

No	Javob	Qiymat
33	*F*	4
34	*E*	5,38
35	*B*	3,6

36–40-savollar

No	To‘g‘ri javob
36	20
37	1,68
38	5,6
39	7,8
40	13,2

41–43-topshiriqlar: javoblar

41-topshiriq.

1) 25 g; 64,4 g
2) 4,11%; 7,29%; 19300 C
3) 1,12 l; 381,4 g
4) 2,57%; 7,45%; 16 g
5) 32,8 ml; 1930 s

42-topshiriq.

1) $CH_3–CH(OH)–CH_3 + CuO \xrightarrow{t} CH_3COCH_3 + Cu + H_2O$ — propanon (4 ball)
2) $CH_3COOCH_3 + NaOH \xrightarrow{t} CH_3COONa + CH_3OH$ — natriy atsetat va metanol (4 ball)
3) $C_6H_{12}O_6 \xrightarrow{\text{spirtli bijg‘ish}} 2C_2H_5OH + 2CO_2$ — etanol (3 ball)
4) $CH_3–C{\equiv}CH + H_2O \xrightarrow{Hg^{2+}} CH_3COCH_3$ — propanon (aseton) (3 ball)
5) $3C_2H_2 \xrightarrow{C,\ t} C_6H_6$ — benzol (4 ball)
6) $C_6H_6 + HNO_3 \xrightarrow{H_2SO_4} C_6H_5NO_2 + H_2O$ — nitrobenzol (3 ball)
7) $n\ CH_2{=}CH–CH{=}CH_2 \rightarrow (–CH_2–CH{=}CH–CH_2–)n$ — polibutadiyen (4 ball)

43-topshiriq.

1) 10,3 g; 7,5 g
2) 1,68 l; 0,15 mol
3) 8 g; 6,35 g
4) 30,55 g
5) 55,7%

Mavzular taqsimoti

No	Mavzu	Tekshiriladigan ko‘nikma	Ball
1	8-sinf; Tirkashev II bob (kimyoviy bog‘lanish)	Bilim va tushunish	1
2	8-sinf (Gey-Lyussak, Avogadro)	Hisoblash va masala yechish	1
3	Tirkashev VI bob (termokimyo, grafik)	Grafik tahlili	1
4	10-sinf §17 (gibridlanish); Abdusamatov	Bilim va tushunish	1
5	Tirkashev *V* bob (OQR, jadval)	Jadval tahlili	1
6	8-sinf (Gey-Lyussak qonuni)	Hisoblash va masala yechish	1

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7	11-sinf 8-bob; Tirkashev *V* bob (elektroliz)	Bilim va tushunish	1
8	11-sinf (kovalent bog‘ turlari)	Rasm bo‘yicha tahlil	1
9	Tirkashev *V* bob (kuchlanish qatori, jadval)	Jadval tahlili	1
10	11-sinf (ekvivalentlar qonuni)	Hisoblash va masala yechish	1
11	Tirkashev I bob; 8-sinf (izoelektron)	Bilim va tushunish	1
12	Tirkashev I bob; 8-sinf (elektron konfiguratsiya)	Bilim va tushunish	1
13	8-sinf; Tirkashev VI bob	Hisoblash va masala yechish	1
14	9-sinf; Tashev (anorganik birikmalar xossalari)	Bilim va tushunish	1
15	Tashev (nitratlar termik parchalanishi)	Hisoblash va masala yechish	1
16	9-sinf (uglerod (IV) oksid)	Hisoblash va masala yechish	1
17	9-sinf; Tirkashev (galogenlar, hisob)	Hisoblash va masala yechish	1
18	9-sinf (ishqoriy metallar)	Hisoblash va masala yechish	1
19	9-sinf	Hisoblash va masala yechish	1
20	10-sinf (nomenklatura §6, §11, §20, §24–26)	Bilim va tushunish	1
21	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
22	10-sinf §15; Abdusamatov	Hisoblash va masala yechish	1
23	10-sinf (spirt, hisob)	Hisoblash va masala yechish	1
24	10-sinf §24; §26; §30 (kumush ko‘zgu)	Hisoblash va masala yechish	1
25	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
26	10-sinf §29 (yog‘larni gidrogenlash)	Hisoblash va masala yechish	1
27	10-sinf (mavzuga oid paragraflar); Abdusamatov	Bilim va tushunish	1
28	10-sinf (ikki bosqichli hisob)	Hisoblash va masala yechish	1
29	10-sinf (pol, hisob)	Hisoblash va masala yechish	1
30	7-sinf (laboratoriya jihozlari, 3-rasm)	Rasm bo‘yicha tahlil	1
31	9-sinf (sifat reaksiyalar)	Bilim va tushunish	1
32	Tirkashev III bob (eritma tayyorlash)	Hisoblash va masala yechish	1
33–35	Aralash masala (moslashtirish)	Ko‘p bosqichli hisob	3
36–40	Qisqa javobli masalalar	Hisoblash	5
41–43	Yozma ish	Usul va hisob (M, A)	75

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